WEEK-9

1.Write a C Program to implement FCFS Disk scheduling algorithm.

AIM:To implement FCFS Disk Scheduling Algorithm and find Total head movement and Average Head movement.

ALGORITHM:

Step 1: Start.

Step 2: Declare the variables `n`, `i`, `head`, and `totalSeek`. Initialize `totalSeek = 0`.

Step 3: Read the number of disk requests `n`.

Step 4: Read the disk requests into the array `requests[]`.

Step 5: Read the initial head position.

Step 6: Display the initial head position as the starting point of the disk request sequence.

Step 7: Repeat the following steps for each disk request from `i = 0` to `n - 1`:

\- Calculate the head movement using:

`abs(requests[i] - head)`

* Add the head movement to `totalSeek`.

* Move the disk head to the current request position.

* Display the current request in the sequence.

Step 8: Calculate the average head movement using:

\\text{Average Head Movement} = \\frac{\\text{Total Head Movement}}{n}

Step 9: Display the disk request sequence, total head movement, and average head movement.

Step 10: Stop.

PROBLEM:

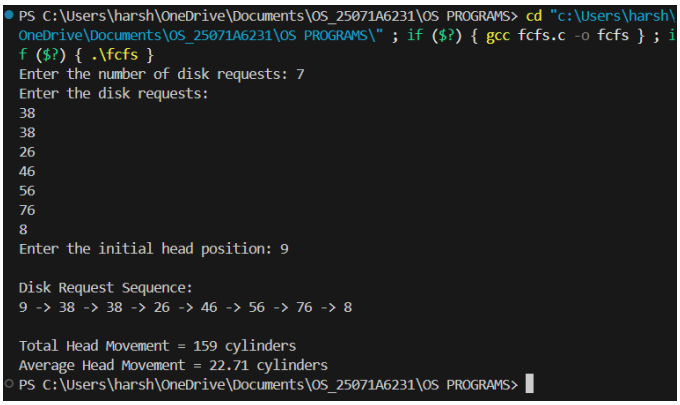
PROGRAM:

```
#include <stdio.h>
#include <stdlib.h>
int main(){
   int n, i;
   int head;
   int totalSeek = 0;
   printf("Enter the number of disk requests: ");
   scanf("%d", &n);
   int requests[n];
   printf("Enter the disk requests:\n");
   for (i = 0; i < n; i++){
      scanf("%d", &requests[i]); }
   printf("Enter the initial head position: ");
   scanf("%d", &head);
   printf("\nDisk Request Sequence:\n");
   printf("%d", head);
   // FCFS: service requests in the same order entered
   for (i = 0; i < n; i++){
      totalSeek += abs(requests[i] - head);
      head = requests[i];
      printf(" -> %d", requests[i]); }
   printf("\n\nTotal Head Movement = %d cylinders", totalSeek);
   printf("\nAverage Head Movement = %.2f cylinders\n",
         (float)totalSeek / n)
```

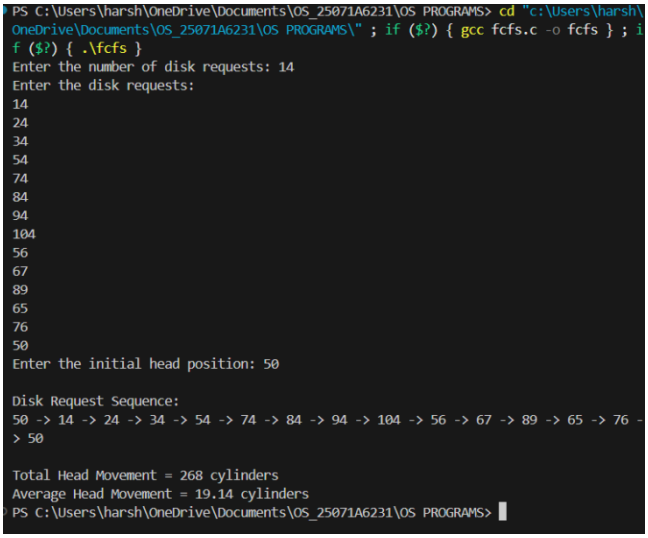
return 0;}

OUTPUT:

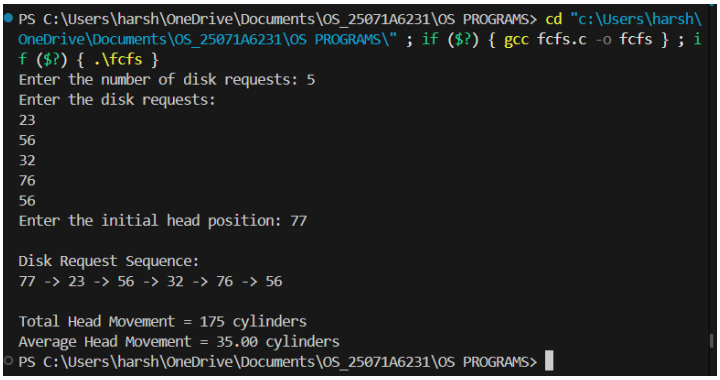
TESTCASE1:



TESTCASE2:



TESTCASE3:



2. Write a C Program to implement SSTF Disk scheduling algorithm.

AIM: To implement Shortest Seek Time First (SSTF)Disk Scheduling Algorithm and find Total head movement and Average Head movement.

ALGORITHM:

Step 1: Start.

Step 2: Declare the variables `n`, `i`, `j`, `head`, `totalSeek`, `min`, `diff`, and `index`. Initialize `totalSeek = 0`.

Step 3: Read the number of disk requests `n`.

Step 4: Read the disk requests into the array `requests[]`.

Step 5: Initialize all elements of the `visited[]` array to `0`, indicating that no request has been serviced.

Step 6: Read the initial head position.

Step 7: Display the initial head position as the starting point of the disk request sequence.

Step 8: Repeat the following steps until all requests are serviced:

* Set `min = 9999` and `index = -1`.

* Compare the current head position with every unvisited disk request.

* Calculate the seek distance using:

`diff = abs(requests[j] - head)`

4. Find the unvisited request with the minimum seek distance.

5. Add the minimum seek distance to `totalSeek`.

6. Move the head to the selected request.

7. Mark the selected request as visited by setting `visited[index] = 1`.

8. Display the selected request in the disk request sequence.

Step 9: Calculate the average head movement.

Step 10: Display the total head movement and average head movement.

Step 11: Stop.

PROBLEM:

VNR VJIET

Name of the Laboratory:

Name of the Experiment:_____

Experiment No:_____ **Date:**_____

PROGRAM:

```
#include <stdio.h>
#include <stdlib.h>
int main(){
    int n, i, j;
    int head, totalSeek = 0;
    int min, diff, index;
    printf("Enter the number of disk requests: ");
    scanf("%d", &n);
    int requests[n];
    int visited[n];
    for (i = 0; i < n; i++) {
        visited[i] = 0;  }
    printf("Enter the disk requests:\n");
    for (i = 0; i < n; i++)  {
        scanf("%d", &requests[i]); }
    printf("Enter the initial head position: ");
    scanf("%d", &head);
    printf("\nDisk Request Sequence:\n");
    printf("%d", head);
    for (i = 0; i < n; i++) {
        min = 9999;
        index = -1;
        for (j = 0; j < n; j++) {
            if (visited[j] == 0)   {
                diff = abs(requests[j] - head);
```

```
            if (diff < min)   {
                min = diff;
                index = j; }
}
}
    totalSeek += min;
    head = requests[index];
    visited[index] = 1;
    printf(" -> %d", head);}
  printf("\n\nTotal Head Movement = %d cylinders",   totalSeek);
  printf("\nAverage Head Movement = %.2f cylinders\n",  (float)totalSeek/n);
  return 0;
}
```

OUTPUT:

TESTCASE 1:

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh\
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc sstf.c -o sstf } ; i
f ($?) { .\sstf }
Enter the number of disk requests: 6
Enter the disk requests:
45
64
34
25
64
65
Enter the initial head position: 55

Disk Request Sequence:
55 -> 64 -> 64 -> 65 -> 45 -> 34 -> 25

Total Head Movement = 50 cylinders
Average Head Movement = 8.33 cylinders
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```

VNR VJIET

Name of the Laboratory:

Name of the Experiment:_____

Experiment No:_____ **Date:**_____

TESTCASE2:

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh\
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc sstf.c -o sstf } ; i
f ($?) { .\sstf }
Enter the number of disk requests: 6
Enter the disk requests:
24
86
7
85
4
3
Enter the initial head position: 100

Disk Request Sequence:
100 -> 86 -> 85 -> 24 -> 7 -> 4 -> 3

Total Head Movement = 97 cylinders
Average Head Movement = 16.17 cylinders
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```

TESTCASE3:

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh\
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc sstf.c -o sstf } ; i
f ($?) { .\sstf }
Enter the number of disk requests: 4
Enter the disk requests:
23
54
23
67
Enter the initial head position: 80

Disk Request Sequence:
80 -> 67 -> 54 -> 23 -> 23

Total Head Movement = 57 cylinders
Average Head Movement = 14.25 cylinders
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```

3.Write a C Prpgram to implement SCAN Disk Scheduling Algorithm.

AIM: To implement SCAN Disk Scheduling Algorithm and find Total head movement and Average Head movement.

ALGORITHM:

1.Start.

2.Read the number of requests, disk size, request queue, initial head position, and direction.

3.Sort the disk requests in ascending order.

4.Display the initial head position.

5.If the direction is right:

Service all requests greater than or equal to the current head position in ascending order.

Move the head to the end of the disk.

Reverse the direction.

Service the remaining requests in descending order.

6.If the direction is left:

Service all requests less than or equal to the current head position in descending order.

Move the head to the beginning of the disk.

Reverse the direction.

Service the remaining requests in ascending order.

7.Calculate the total head movement.

8.Calculate the average head movement by dividing the total by the number of requests.

9.Display the results.

10.Stop.

PROBLEM:

VNR VJIET

Name of the Laboratory: _____

Name of the Experiment: _____

Experiment No: _____ **Date:** _____

PROGRAM:

```
#include <stdio.h>
#include <stdlib.h>
void sort(int a[], int n){
    int i, j, temp;
    for (i = 0; i < n - 1; i++)   {
        for (j = 0; j < n - i - 1; j++)      {
            if (a[j] > a[j + 1]) {
                temp = a[j];
                a[j] = a[j + 1];
               a[j + 1] = temp;   }}}}
int main()
{
    int n, diskSize, head, direction;
    int i, j, totalSeek = 0;
    int requests[100];
    printf("Enter the number of disk requests: ");
    scanf("%d", &n);
    printf("Enter the disk size: ");
    scanf("%d", &diskSize);
    printf("Enter the disk requests:\n");
    for (i = 0; i < n; i++) {
        scanf("%d", &requests[i]);    }
  printf("Enter the initial head position: ");
    scanf("%d", &head);
    printf("Enter direction (1 for right, 0 for left): ");
    scanf("%d", &direction);
```

```
sort(requests, n);
    printf("\nDisk Request Sequence:\n");
    printf("%d", head);
    if (direction == 1)  {
       for (i = 0; i < n; i++)   {
          if (requests[i] >= head) {
             totalSeek += abs(requests[i] - head);
             head = requests[i];
             printf(" -> %d", head);}}
       if (head != diskSize – 1) {
          totalSeek += abs((diskSize - 1) - head);
          head = diskSize - 1;
          printf(" -> %d", head);}
       for (i = n - 1; i >= 0; i--)   {
          if (requests[i] < head) {
             totalSeek += abs(requests[i] - head);
             head = requests[i];
             printf(" -> %d", head);}}
    else  {
       for (i = n - 1; i >= 0; i--){
        if (requests[i] <= head){
             totalSeek += abs(requests[i] - head);
             head = requests[i];
             printf(" -> %d", head);  }}
       if (head != 0){
          totalSeek += head
```

```
    head = 0;
            printf(" -> %d", head);}
        for (i = 0; i < n; i++){
            if (requests[i] > head){
                totalSeek += abs(requests[i] - head);
                head = requests[i];
                printf(" -> %d", head);}}}
    printf("\n\nTotal Head Movement = %d cylinders",
            totalSeek);
    printf("\nAverage Head Movement = %.2f cylinders\n",
            (float)totalSeek / n);
    return 0;
}
```

OUTPUT:

TESTCASE1:

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh\
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc scan.c -o scan } ; i
f ($?) { .\scan }
Enter the number of disk requests: 6
Enter the disk requests:
12
26
56
57
36
29
Enter the initial head position: 50
Enter direction (1 for right, 0 for left): 1

Disk Request Sequence:
50 -> 56 -> 57 -> 56 -> 36 -> 29 -> 26 -> 12

Total Head Movement = 52 cylinders
Average Head Movement = 8.67 cylinders
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```

TEST CASE2:

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh\
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc scan.c -o scan } ; i
f ($?) { .\scan }
Enter the number of disk requests: 5
Enter the disk requests:
45
65
58
26
25
Enter the initial head position: 25
Enter direction (1 for right, 0 for left): 1

Disk Request Sequence:
25 -> 25 -> 26 -> 45 -> 58 -> 65 -> 58 -> 45 -> 26 -> 25

Total Head Movement = 80 cylinders
Average Head Movement = 16.00 cylinders
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```

TESTCASE3:

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh\
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc scan.c -o scan } ; i
f ($?) { .\scan }
Enter the number of disk requests: 5
Enter the disk requests:
54
63
89
65
23
Enter the initial head position: 25
Enter direction (1 for right, 0 for left): 1

Disk Request Sequence:
25 -> 54 -> 63 -> 65 -> 89 -> 65 -> 63 -> 54 -> 23

Total Head Movement = 130 cylinders
Average Head Movement = 26.00 cylinders
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```